ANORD MARDIX
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MDX Switchgear

CATALOGUE



IEC 61439-2

MDX LOW-VOLTAGE SWITCHGEAR SOLUTIONS

Why MDX?

MDX low-voltage switchgear is welded modular distribution assemblies, tested to IEC 61439-2 and designed to deliver safety, uptime, and serviceability in data centres, colocation facilities, commercial, industrial and other critical infrastructure applications.

High uptime

MDX switchgear are designed for continuous operation using a breaker-centric protection architecture. Integration of IEC 60947-2 ACBs and MCCBs, combined with a fully rated bus system and short-time withstand capability up to 100kA for 1 second, ensures predictable fault behaviour. Selective coordination limits outages to affected circuits, keeping critical loads energized and reducing recovery time.

Compact density

High current ratings and optimized circuit breaker stacking allow MDX switchgear to deliver dense power distribution in a compact footprint. Busbars rated up to 1200A to 6300A, standardised section widths, and front access to instrumentation maximise feeder capacity while minimising electrical room space and installation constraints.

Proven performance

MDX switchgear is fully assembled and tested as a complete system in accordance with IEC 61439-2, verifying thermal performance, dielectric integrity, and short-circuit withstand capability. Seismically qualified configurations are available where required. This comprehensive testing helps validate real-world performance while reducing commissioning effort and operational risk.

Design flexibility

MDX switchgear supports modular, standardised construction, enabling flexible design configurations aligned with customer requirements. Circuit breaker and meter agnostic integration, together with support for Modbus TCP/IP and Modbus RTU, allows seamless integration with customer monitoring and control systems while maintaining compatibility with established standards.

Digital ready

MDX switchgear incorporates high-accuracy metering that integrates seamlessly with the data centre BMS, enabling remote, real-time visibility into power quality and operating parameters. HMIs and pilot device indicators also provide clear local status indication, supporting confident and reliable day to day operation.

Data center ready

MDX switchgear is rated up to 690V AC and features natural convection cooling, with indoor construction compliant with IEC 61439-2. Purpose-built for data centres and other critical facilities, MDX switchgear offers front and front / rear access operation, scalable section design, and robust protection features to support long term reliability, operational efficiency, and future expansion.

Benefits of MDX switchgear



Construction and design

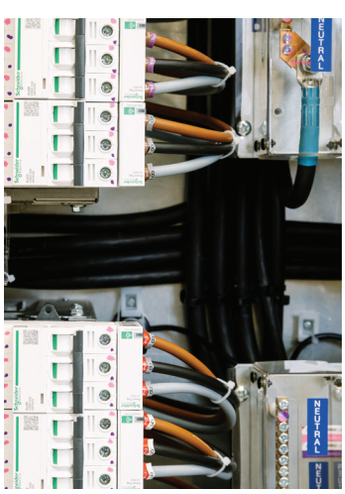
- Modular, welded construction supports scalability and future expansion
- Copper busbars rated up to 6300A enable reliable current distribution
- Compact enclosure design helps minimise electrical room space requirements
- Optimised circuit breaker stacking maximises feeder density within each section



Safety and protection

- IP2X construction helps prevent inadvertent contact with live parts during operation and maintenance
- Segregated compartments limit fault propagation and enhance personnel safety
- Guarded MCCB terminals support safe isolation of energised connections
- Circuit breaker based selective coordination isolates faults while maintaining continuity to critical loads
- Energy Reduction Maintenance Setting (ERMS) can be implemented to temporarily reduce incident energy levels during maintenance



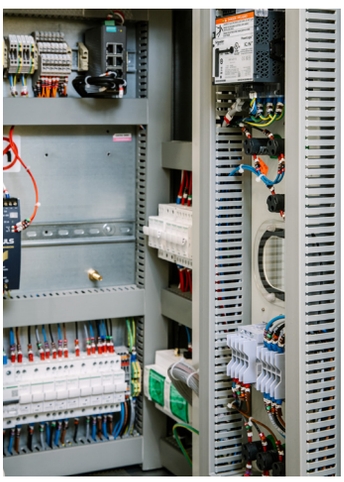


Operation and maintenance

- Clear ON, OFF, and TRIP indications improve operator awareness and response
- LOTO ready circuit breaker mechanisms support safe and compliant maintenance procedures
- Mechanical and electrical key interlocks help prevent unsafe operating sequences

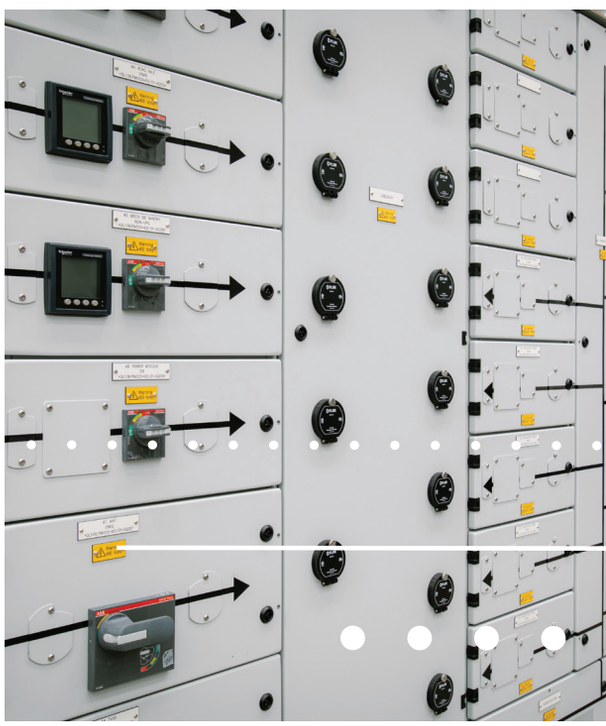
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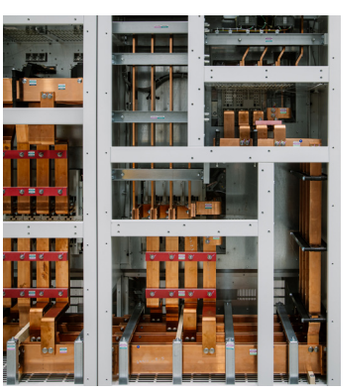
Benefits of MDX switchgear



Digital readiness and monitoring

- Meter agnostic architecture supports flexible monitoring configurations
- Support for Modbus TCP/IP and Modbus RTU enables integration with BMS and EMS platforms
- Customer defined metering points provide targeted system visibility
- Front accessible instrumentation allows safe monitoring and interaction in live environments





Reliability and uptime

- MDX switchgear design ensures predictable performance
- High interrupting capacity breakers enable fast and reliable fault clearing
- Natural convection cooling reduces system complexity and ongoing maintenance needs
- Design supports preventive maintenance required for critical environments



Certification and testing

- Assemblies tested/verified to IEC 61439-2
- Busbars verified for short-time withstand up to 100kA for 1 second
- Seismic options available for regulated or high-risk installations
- Manufacturing conducted under ISO 9001 certified systems

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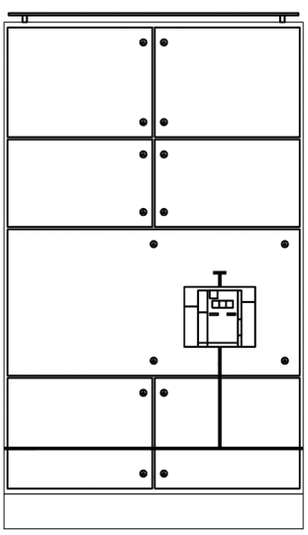
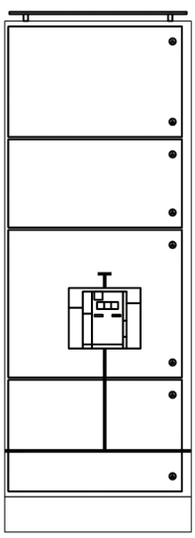
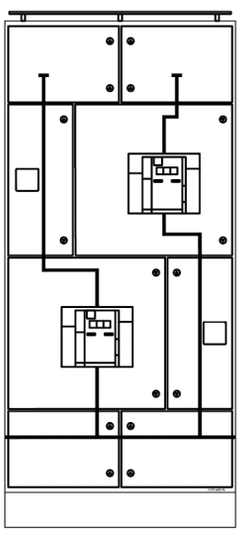
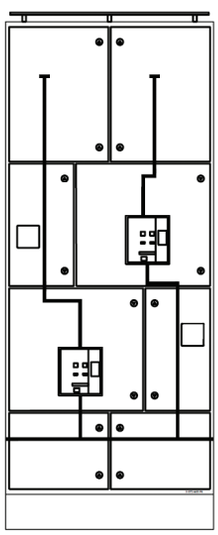
Technical specifications

Parameter	Specification
Product	
Product	MDX low-voltage switchgear
Target markets	Data centres, colocation facilities, critical infrastructure, commercial and industrial
Colour	Epoxy paint RAL 7035 (other colours available on request)
Quality management	Manufactured under ISO 9001 certified quality management system
Electrical	
Rated operational voltage (Ue)	Up to 690V AC
Rated insulation voltage (Ui)	Up to 1000V
Rated impulse withstand voltage (Uimp)	8–12kV
Rated frequency	50Hz
Peak withstand current (Ipk)	Up to 220kA
Short-time withstand current (Icw)	50kA for 3s, up to 100kA for 1s
Conditional short-circuit current (Ipk)	Up to 220kA
Main busbar rating	Up to 8000A
Vertical busbar (feeder distribution)	Up to 5000A
Neutral busbar rating	100–200%
Incomer ACB ratings	800–6300A
Construction	
Construction type	Cubical construction; 2mm galvanised Zintec steel, fully welded and powder-coated
Busbar material	High-density 99% pure copper (tin-plated copper optional)
Cooling method	Natural convection with top and bottom vents integrated into panel design
Panel width	Modular cubicles, 600–1300mm
Internal separation	Up to Form 4b type 7, per IEC 61439-2
Mounting type	Fixed or withdrawable
Cable entry	Top or bottom, with gland plates
Incomer panel	800–6300A ACB
Bus coupler panel	800–6300A ACB
Outgoing feeder panels	160–4000A MCCB
ACB stacking	Up to 2 ACBs per column (maximum 2000A per device)
MCCB configuration (single column)	8 × 250A or 6 × 630A, 4-pole (TP&N)
MCCB configuration (double column)	16 × 250A or 12 × 630A, 4-pole (TP&N)

Technical specifications

Safety	
Personnel protection	IP2X internal protection; touch-safe instrumentation
Interlocking	Electrical and mechanical key interlocking systems
Circuit breaker safety	Shutters, terminal shields, and mechanical interlocks (where applicable)
Thermal monitoring	Non-invasive thermal imaging ports for proactive maintenance
Environment	
Degree of protection	IP31, indoor industrial
Ambient temperature	Daily average 35°C, peak 40°C
Storage temperature	0°C to +50°C
Relative humidity	≤ 90%, non-condensing
Maximum altitude	1000m above sea level
Control and Communication	
Communication protocols	Modbus TCP/IP, Modbus RTU
System integration	BMS, EMS, and monitoring systems (per customer specification)
Front-face operation	HMI interface, push buttons, and operating handles
Compliance and Testing	
Design verification	Assemblies tested/verified to IEC 61439-2
Seismic qualification	Seismic tested to IEEE Std 344-2013, SDS 1g
Optional testing	Internal arc fault testing to IEC 61641 (class A, B, C, I)

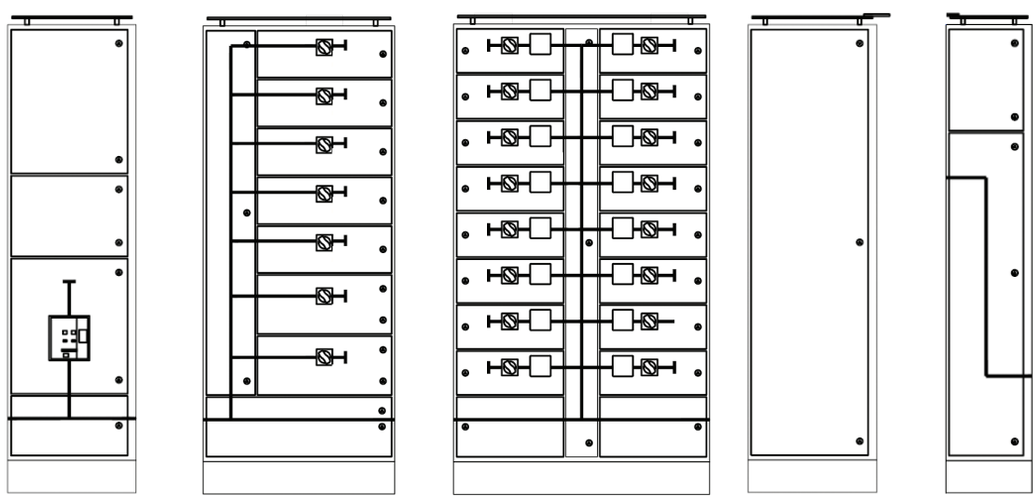
Standard cabinet configurations

4000A - 6300A Single stack	**800A - 4000A** Single stack	**800A - 2000A** Double stack	**800 - 1250A** Double stack
			

Cabinet type	4Ph/4W width (mm)	Depth (mm)	Height (mm)	# of devices	Voltage	Device type	Main busbar rating	Cable entry/exit
4000A - 6300A	1300	800 or 1400	2200	Single ACB	Up to 690V AC	Fixed or draw out	Thermally rated	Top or bottom
800A - 4000A	800	800 or 1400	2200	Single ACB				
800A - 2000A	1000	800 or 1400	2200	Double ACB				
800A - 1250A	900	800 or 1400	2200	Double ACB / MCCB				

Standard cabinet configurations

630A - 1600A ACB/MCCB	100A - 250A MCCB Single stack	100A - 630A MCCB Double stack	Cableway	Transition



Cabinet type	4Ph/4W width (mm)	Depth (mm)	Height (mm)	Voltage	Device type	Number of devices	Cable entry/exit
630A - 1600A ACB/MCCB	600	800 or 1400	2200	Up to 690V AC	Fixed / withdrawable	1 device only	Top or bottom
100A - 250A Single stack	900	800 or 1400	2200	-	Fixed / plug-in	8 × 250A or 6 × 630A, 4-pole (TP&N)	-
100A – 630A Double stack	1400	800 or 1400	2200	-	Fixed / plug-in	16 × 250A or 12 × 630A, 4-pole (TP&N)	-
Cableway	600	800 or 1400	2200	-	-	-	-
Transition	400	800 or 1400	2200	-	-	-	-

Services and lifecycle support

Anord Mardix provides structured services to support MDX switchgear throughout the project lifecycle. Services are delivered by experienced field and engineering teams to ensure correct installation, verified performance, and smooth handover to operations.

1. Project and pre installation support

Includes technical review against project one-line diagrams and site requirements, installation guidance for layout, clearances, cable entry, environmental conditions, and provision of drawings, datasheets, and test records for submittal and site use.

2. Testing and verification

Includes verification of protection settings against the approved coordination study, operational checks of interlocks, and safety features, and validation of Modbus TCP/IP and Modbus RTU communications where provided.

3. Commissioning and startup

On-site startup support is provided to verify installation readiness for energisation, followed by functional testing of circuit breakers, interlocks, instrumentation, and control circuits, and integrated testing with upstream and downstream systems prior to handover.

4. Training and handover

Training covers normal operation, circuit breaker handling, safety features, isolation procedures, racking operations, and inspection practices, supported by a review of handover documentation for ongoing operation.

5. After sales support

After commissioning, support includes technical assistance for operational questions, guidance on spare parts, and field service support for troubleshooting or modifications as required.

6. Warranty

A standard 12-month defects warranty is provided. Extended warranty options are available subject to scope and agreement.

Appendix

Application considerations and definitions

Air Circuit Breaker (ACB)

A high capacity power circuit breaker is integrated into the MDX switchgear as an incoming, bus-coupler, or large feeder device. In segregated compartments, and engineered to support system wide selectivity, short-time withstand, and safe maintenance isolation.

MDX switchgear

A modular, low-voltage power distribution assembly to IEC 61439-2. MDX switchgear is designed for high-density, critical applications and integrates ACBs, MCCBs, busbars, protection, and control systems into a scalable, front-access enclosure.

Architecture design

An MDX switchgear design philosophy that allows integration of approved breakers from multiple manufacturers without changing the core switchgeararchitecture. This enables flexibility in protection strategy, supply chain resilience, and customer standardisation.

Breaker-based protection

A protection architecture in which fault interruption, isolation, and coordination are achieved primarily through the characteristics and settings of ACBs and MCCBs integrated into the MDX switchgear.

Busbar

The primary current carrying copper conductors in the MDX switchgear that distribute power across sections. Front and rear, mechanically braced, and rated from 800A up to 6300A to support high density power distribution.

Closed door operation

An MDX switchgear safety feature that allows circuit breaker operation and status verification with compartment doors closed, maintaining the integrity of dead-front protection and reducing operator exposure during energised operation.

IP2X construction

A fundamental MDX switchgear safety design in which live electrical parts are fully enclosed behind barriers and covers. This ensures operators cannot contact energised components during normal operation or routine maintenance.

Drawout ACB

A configuration in which the ACB is mounted on a racking mechanism within the MDX switchgear, allowing it to be moved between connected, test, and disconnected positions. This enables safe isolation, inspection, and maintenance without disturbing upstream busbars.

Front/rear access serviceable

An MDX switchgear layout feature in which all normal operation, metering, circuit breaker service, and preventive maintenance activities are performed from the front or rear of the switchboard.

Fully rated system

The MDX switchgear system employs a fully rated design methodology in which all circuit breakers and busbars operate at their current ratings, or applicable diversity factors. This ensures consistent thermal behaviour and reliable, predictable protection performance under all operating conditions.

Ingress protection (IEC 60529)

The enclosure rating of the MDX switchgear, intended for indoor installations only. Forms of separation construction protects against accidental contact with live parts and falling debris, IP2X ensures finger safe design.

Interlocking (electro/mechanical and key)

A mechanical key based interlocking system integrated into MDX switchgear to ensure safe operating sequences, such as preventing simultaneous connection of multiple sources or access to compartments under unsafe conditions.

Appendix

Application considerations and definitions

LOTO ready isolation

An MDX switchgear safety feature that provides lockout/tagout capability at circuit breaker operating mechanisms, allowing maintenance personnel to secure isolation points in compliance with site safety procedures.

Moulded Case Circuit Breaker (MCCB)

A compact IEC 60947 MCCB used in the MDX switchgear for feeder and branch circuit protection. MCCBs are fixed or plug-in and are arranged to maximize feeder compactness while maintaining IP rating.

Selective coordination

A protection strategy implemented within the MDX switchgear whereby time-current characteristics are selected so that only the circuit breaker closest to a fault operates, maintaining power to unaffected sections

Short-circuit withstand rating

The verified ability of the MDX switchgear, including busbars and supports, to safely withstand and carry high fault currents.

Touch safe instrumentation

Instrumentation and control devices in the MDX switchgear that are installed behind protective barriers or covers, allowing interaction without exposure to energised components.

Fully built and tested assembly

Fully built and tested to IEC 61439-2, through rigorous testing these assemblies ensure safety, reliability, and consistent performance regarding temperature rise, short-circuit withstand, and insulation, making them ideal for critical infrastructure.

Maintenance isolation

Isolation provided by drawout ACBs and plug-in MCCB's in MDX switchgear, to physically isolate a breaker from live busbars while maintaining enclosure integrity, enabling safer maintenance.

Static Transfer Switch (STS) integration

STS integration in MDX switchgear provides fast, controlled, and automatic transfer between power sources, ensuring resilient and uninterrupted supply to critical loads within a compact, reliable assembly.

Active Harmonic Filter (AHF)

AHF integration in MDX switchgear ensures clean, compliant power for data centres by dynamically mitigating harmonics, improving energy efficiency, and protecting critical IT loads and infrastructure.

Packaged substations

Transformer packaged substations align directly with MDX switchgear, eliminating redundant enclosures and cabling while using optimized, factory-engineered layouts to reduce space. Pre-integrated, tested connections shorten installation time, reduce losses, and lower total installed cost while improving overall efficiency.

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Anord Mardix, a Flex company, is a leading provider of power distribution and protection equipment to the global market. We are a vendor of choice to many of the world's core markets, from data centers to cloud computing.

With over 100 years of combined expertise we offer clients the most comprehensive product range in the industry. Our ability to customize our product offering to each project, and manufacture on a global scale, ensures reliable and bespoke end-to-end critical power solutions to all our customers, from independent providers to hyperscale leaders.

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04-2026EN

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